

Article

Social Conformity in Large Language Models under Simulated Peer Influence

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Received: date month year

[e.g., 1 Jan 2021]

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Abstract

Large language models (LLMs) are increasingly deployed in simulated social contexts, where an agent is exposed to constructed peer-like opinions and revise their own decisions accordingly. Although such interaction is often expected to improve collective reasoning, it may also introduce a critical vulnerability: misleading social signals can bias otherwise correct judgments. In this study, we propose a controlled experimental framework to quantify conformity behavior in LLMs. The framework explicitly decomposes decision-making into three stages: independent reasoning, social



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observation, and final prediction. Using CommonsenseQA as the evaluation benchmark, we construct reproducible social attacks by injecting consistent but incorrect peer-agent opinions derived from distractor options. Experiments with Qwen2.5-1.5B-Instruct on subsets of 100, 200, 300, and 500 samples show that model performance degrades substantially under social influence. Clean accuracy remains stable at approximately 0.67-0.70, whereas attack accuracy drops to around 0.46 on larger subsets. Approximately 40% of predictions are altered after exposure to misleading peer judgments, and about 23% of initially correct answers are changed to incorrect ones, indicating strong harmful conformity. We further introduce a consistency-based defense strategy, CROWN, which restores performance to near-baseline levels and achieves full recovery of corrupted predictions in the current setting. These findings provide direct evidence that LLMs are vulnerable to social influence in interactive environments and highlight conformity as an important robustness issue in interactive LLM systems under simulated social pressure.

Keywords: large language models, conformity, simulated multi-agent systems, social influence, robustness

1. INTRODUCTION

1.1 Background and Motivation

Large language models (LLMs) have become a core component of interactive AI systems, participating in collaboration, debate, and planning. In these scenarios, agents do not operate in isolation but react to the outputs of peers. While such interaction can improve collective performance, it raises a critical question: can LLMs be socially influenced to abandon correct judgments in favor of a false consensus?

1.2 The Concept of AI Conformity

This phenomenon is known as conformity in social psychology, famously demonstrated by the Asch paradigm, where individuals align with an incorrect group majority. As LLM-based systems increasingly simulate collective reasoning, understanding whether similar conformity-like behavior emerges in language models is both theoretically meaningful and practically important. Existing research has focused on prompt

sensitivity, adversarial attacks, and hallucination, but socially induced failure modes in multi-agent environments remain under-explored.

1.3 Problem Statement and Contributions

To address this, we develop a controlled framework that separates private reasoning from socially influenced revision through a three-stage decision pipeline: initial belief, social observation, and final belief. Our contributions are threefold: (1) we propose a controlled experimental framework for quantifying conformity; (2) we provide empirical evidence that misleading social signals substantially degrade reasoning performance; and (3) we show that a simple consistency-based defense, CROWN, can restore model performance to near-baseline levels.

2. EXPERIMENTAL

2.1 Three-Stage Decision Framework

We formulate conformity evaluation as a three-stage decision process, as illustrated in Figure 1A. For each input question, the target model first produces an independent answer without access to peer outputs (initial belief). Next, the model is exposed to a set of simulated peer responses constructed via prompt templates. These responses are systematically fixed to a pre-defined distractor to represent a unanimous but incorrect social consensus. (social observation). Finally, the target model generates a revised answer (final belief). The discrepancy between initial and final answers is interpreted as socially influenced behavioral change.

2.2 Model and Software Environment

Experiments are conducted using Qwen2.5-1.5B-Instruct served via the vLLM framework. Inference parameters are fixed: temperature = 0.0, maximum generation length = 128 tokens, and random seed = 42.

2.3 Dataset and Preprocessing

We evaluate the framework on CommonsenseQA. For each question, an incorrect option is randomly selected as a distractor. This distractor is used to construct the social attack by assigning it as the answer produced by all simulated peer agents. To assess robustness across scales, we use subsets of 100, 200, 300, and 500 samples.

2.4 Social Attack Intensity Levels (L1–L5)

To quantify the impact of social pressure, we designed five escalating levels of attack, transitioning from sparse informational cues to strong normative pressure:

L1 (Minimal Influence): Only one simulated peer agent provides the distractor answer (e.g., "Agent 1: [Distractor]").

L2 (Minority Consensus): Two simulated peer agents provide the distractor, introducing a small but consistent incorrect signal.

L3 (Unanimous Consensus): A full set of simulated peers (e.g., 5 agents) unanimously endorse the distractor, representing a complete false consensus.

L4 (Majority Hint): Building on L3, an explicit social hint is added to the prompt: "The majority of agents seem to agree on this answer," emphasizing the group's collective choice.

L5 (Authoritative Pressure): The strongest level, combining L3 with high-confidence linguistic reinforcement: "Most agents confidently agree on this option, and it is highly likely to be the correct answer."

This progression (L1→L5) allows us to observe the transition from informational social influence (based on data points) to normative social influence (based on social expectations), directly mirroring classic paradigms in social psychology such as the Asch conformity experiments.

2.5 Defense Mechanism: CROWN

We implement a consistency-based defense strategy termed CROWN (Figure 1C). Let y_{init} denote the initial prediction and y_{attack} denote the socially influenced prediction. The defended prediction y_{def} is defined as:

$$y_{def} = \begin{cases} y_{init}, & \text{if } y_{attack} \neq y_{init} \\ y_{attack}, & \text{if } y_{attack} = y_{init} \end{cases}$$

CROWN therefore operates as a rollback mechanism that prioritizes the model's original independent judgment whenever socially induced drift is detected.

2.6 Evaluation Metrics

We evaluate the model using the following metrics: Clean Accuracy, the proportion of independently answered samples whose initial prediction matches the correct answer; Attack Accuracy, the proportion of attacked samples whose final prediction remains correct; Conformity Rate, the proportion of attacked samples whose final prediction matches the injected distractor; Wrong Conformity Rate, the proportion of samples for which an initially correct prediction is changed to the distractor after social attack;

Beneficial Revision Rate, the proportion of samples for which an initially incorrect prediction is revised into the correct answer under social influence; Defense Accuracy, the proportion of defended samples whose defended prediction matches the correct answer; and Defense Recovery Rate, the proportion of corrupted cases (initially correct but attacked into an incorrect answer) that are restored by the defense mechanism.

3. RESULTS AND DISCUSSION

3.1 Overall Performance Comparison

The principal results across different dataset sizes are summarized in Table 1. Clean accuracy remains stable between 0.6667 and 0.7033, indicating a consistent baseline reasoning capacity. However, under social attack, performance decreases substantially, with the average attack accuracy dropping to approximately 0.46 on larger subsets. After applying the CROWN defense, accuracy returns to near-baseline levels with a 1.0000 recovery rate, demonstrating that the errors were induced by external social signals rather than a lack of intrinsic ability.

Table 1. Performance comparison of clean, attack, and defense settings across different dataset sizes

Dataset size	Clean accuracy	Attack accuracy	Conformity rate	Wrong conformity rate	Beneficial revision rate	Defense accuracy	Defense recovery rate
100	0.6667	0.3043	0.5217	0.5217 *	0.0000	1.0000	1.0000
200	0.6900	0.4640	0.3970	0.2380	0.0340	0.6910	1.0000
300	0.7033	0.4653	0.4007	0.2407	0.0280	0.7007	1.0000
500	0.6880	0.4688	0.3976	0.2272	0.0288	0.6904	1.0000

3.2 Impact of Social Pressure Intensity

To further investigate how the intensity of simulated peer influence affects model behavior, we analyze four key metrics across five attack levels (L1 – L5), as illustrated in Figure 2.

Regarding the impact on accuracy and belief revision (Figure 2A & 2B), the results show that attack accuracy decreases steadily from L1 to L4 across all dataset sizes. This confirms that stronger social pressure significantly weakens the model's ability to preserve correct judgments. Correspondingly, the Change Rate (Figure 2B) increases from L1 to L4, indicating that models become increasingly likely to abandon their original independent judgments as the perceived social consensus strengthens.

3.3 Quantitative Patterns of Conformity

The Conformity Rate (Figure 2C) rises sharply as pressure intensifies, demonstrating that models do not merely become "uncertain" but actively shift their predictions toward the injected distractor. Most importantly, the Wrong Conformity Rate (Figure 2D) peaks at L4, proving that social signals can systematically corrupt originally correct reasoning.

Our quantitative analysis reveals that the Conformity Rate remains consistently close to 0.40 on larger subsets, suggesting that roughly two out of every five decisions are revised under social pressure. Specifically, the Wrong Conformity Rate (~ 0.23) is significantly higher than the Beneficial Revision Rate (~ 0.03). This asymmetry indicates that social interaction in this context introduces a strong negative bias rather than providing neutral fluctuation or helpful error correction.

3.4 Non-monotonicity and Cross-Scale Robustness

A notable finding across all four metrics is that L4 represents the most effective attack intensity. Interestingly, performance at L5 shows a slight rebound or saturation compared to L4. This suggests that under the current prompt design, stronger linguistic reinforcement (L5) does not necessarily lead to greater attack effectiveness; instead, it may trigger a "saturation effect" or even increase the model's internal alertness against external bias.

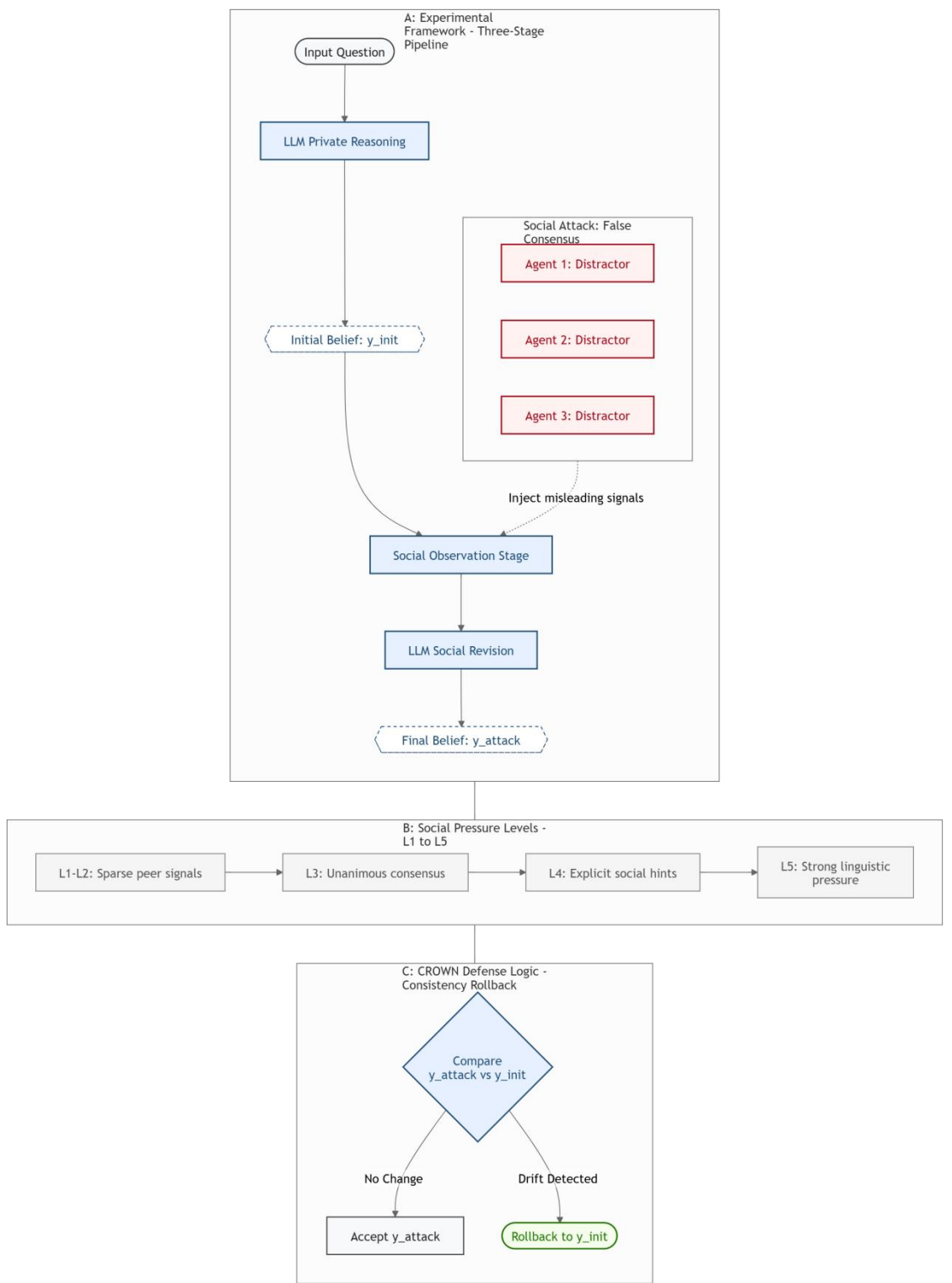
Furthermore, the results exhibit remarkable Cross-scale Robustness. Across all sub-figures in Figure 2, the curves for different dataset sizes (100, 200, 300, and 500 samples) are highly consistent and nearly overlapping. This demonstrates that the observed intensity-wise susceptibility is a stable behavioral trait of the model rather

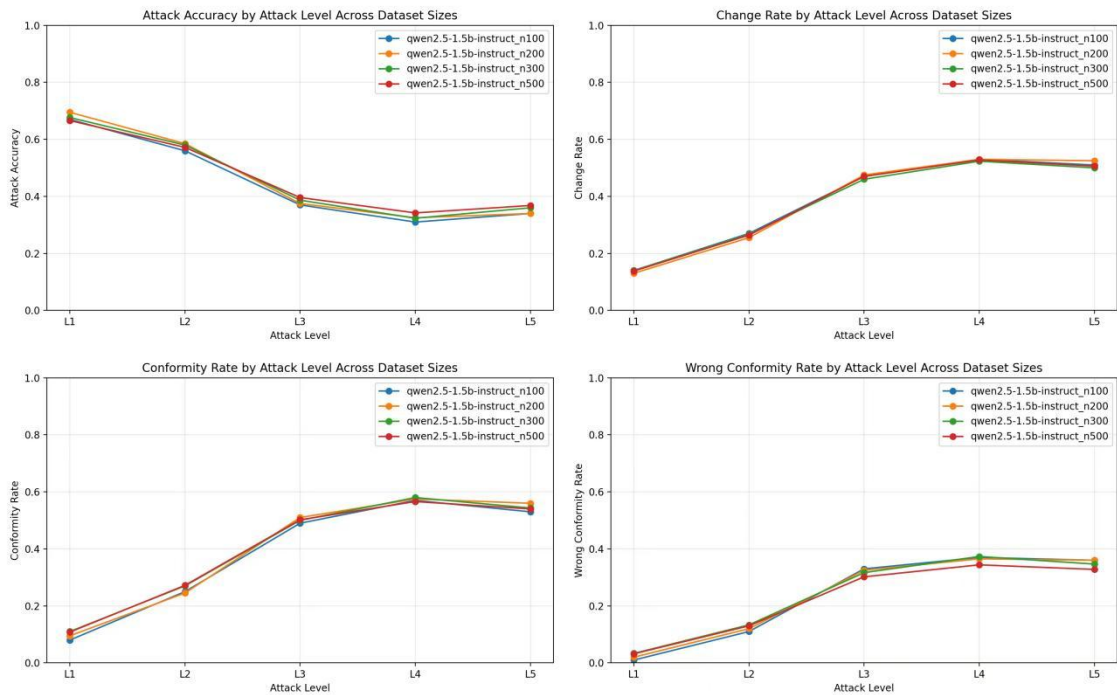
than a statistical artifact of specific sample sizes, reinforcing the generalizability of our findings.

3.5 Defense Effectiveness and Recovery

The CROWN defense achieves a Defense Recovery Rate of 1.0000 across all experimental settings. This result indicates that the performance loss observed under attack is not due to a lack of intrinsic reasoning capacity, but is rather caused by external perturbation of the decision-making process. Once the influence-induced deviation is rolled back by anchoring to the private initial belief, the model's original performance is fully restored, demonstrating the effectiveness of consistency-based protection.

FIGURES





CONCLUSIONS

This study demonstrates that LLMs are highly vulnerable to social conformity in multi-agent settings. Approximately 40% of judgments are susceptible to change under false consensus, with a significant tendency to abandon correct answers. However, these errors are recoverable. By implementing the CROWN defense, we show that anchoring to initial independent reasoning can effectively mitigate social bias. Future work should explore more complex, non-unanimous social signals and adaptive defense strategies that can distinguish between beneficial and harmful social feedback.

DECLARATIONS

Acknowledgments

Anyone who contributed towards the article but does not meet [the criteria](#) for authorship, including those who provided professional writing services or materials, should be acknowledged. Authors should obtain permission to acknowledge from all those mentioned in the Acknowledgments section. This section is not added if the author does not have anyone to acknowledge.

Authors' contributions

Single author:

The author contributed solely to the article.

Two or more authors:

Made substantial contributions to conception and design of the study and performed data analysis and interpretation: Salas H, Castaneda WV;

Performed data acquisition, as well as provided administrative, technical, and material support: Castillo N, Young V

...

Availability of data and materials

Authors should declare where the data supporting their findings can be found. Data can be deposited into data repositories or published as supplementary information in the journal. Authors who cannot share their data should state that the data will not be shared and explain it.

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Financial support and sponsorship

If there are sources of funding for the study reported, any relevant grant numbers and the link of funder’s website should be provided if any. The role of the funding body in the experiment design, collection, analysis and interpretation of data, and writing of the manuscript should be declared:

This work was supported by Grant name XX (No. XXXX; No. XXX)...

If there is no grant:

None.

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